

HCI & CG — Lab 1

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Short Answer

Why does adding an Alpha channel (RGBA) increase an image array's memory consumption by 33% compared to standard RGB?

A standard RGB image stores 3 channels per pixel (Red, Green, Blue), each typically using 1 byte (8 bits) in an `np.uint8` array, so every pixel occupies 3 bytes. Adding an Alpha channel for transparency introduces a 4th channel of the same bit depth, so each pixel now occupies 4 bytes instead of 3.

The increase in memory is therefore the ratio of the extra channel to the original channel count: 1 additional channel out of the original 3 channels equals $1/3$, or approximately 33.33%. This can also be seen directly from the array shape: an $(H, W, 3)$ array has $H \times W \times 3$ elements, while an $(H, W, 4)$ array has $H \times W \times 4$ elements — a $4/3$ ratio, i.e., a 33% increase in total elements and total bytes (`.nbytes`), even though no additional color information is being stored, only per-pixel opacity/transparency data.